

Measures of soil water content for a volume of soil with dimensions 10 cm × 10 cm × 10 cm that has a mass of 1,700 g when wet, 1,450 g when oven-dry, and a particle density ρ_s of 2.65 g/cm³.

Quantity	Amount
Mass of water	$M_w = 1,700 \text{ g} - 1,450 \text{ g} = 250 \text{ g}$
Mass of soil solids	$M_s = 1,450 \text{ g}$
Bulk volume of soil	$V_t = 10 \text{ cm} \times 10 \text{ cm} \times 10 \text{ cm} = 1,000 \text{ cm}^3$
Bulk density	$\rho_b = M_s/V_t = 1.45 \text{ g/cm}^3$
Porosity	$\theta_{sat} = 1 - \rho_b/\rho_s = 0.453$
Volume of water	$V_w = M_w/\rho_w = 0.25 \text{ cm}^3$
Gravimetric water content	$\theta_g = M_w/M_s = 0.172 \text{ g/g}$
Volumetric water content	$\theta_v = \theta_g \times \rho_b/\rho_w = 0.25 \text{ cm}^3/\text{cm}^3$
Degree of saturation*	$S = \theta_v/\theta_{sat} = 0.552$
Depth of water	$\theta_v \times \Delta z = 0.25 \times 10 \text{ cm} = 2.5 \text{ cm}$
Mass of water per unit area**	$W = M_w/\Delta z \times \Delta z = 250 \text{ g}/10 \text{ cm} \times 10 \text{ cm} = 2.5 \text{ g/cm}^2$
Volume of water per unit area**	$V = V_w/\Delta z \times \Delta z = 0.25 \text{ cm}^3/10 \text{ cm} \times 10 \text{ cm} = 0.0025 \text{ cm}^3/\text{cm}^2$

*also called 'water-filled pore space'

** per centimeter depth